

HYBRID INTELLIGENT OBJECT DETECTION SYSTEM

NLP Project Report|Chandigarh University

1. Abstract

This project presents a Hybrid Intelligent Object Detection System that combines YOLO for real-time detection with Generative AI (Google Gemini) for intelligent verification. YOLO ensures fast detection, while Gemini improves accuracy by refining low-confidence predictions. The system enhances robustness and maintains real-time performance using asynchronous processing.

Introduction

Object detection is a critical task in computer vision with applications in surveillance, automation, and smart systems. Traditional models like YOLO are fast but limited to trained classes.

This project introduces a hybrid approach:

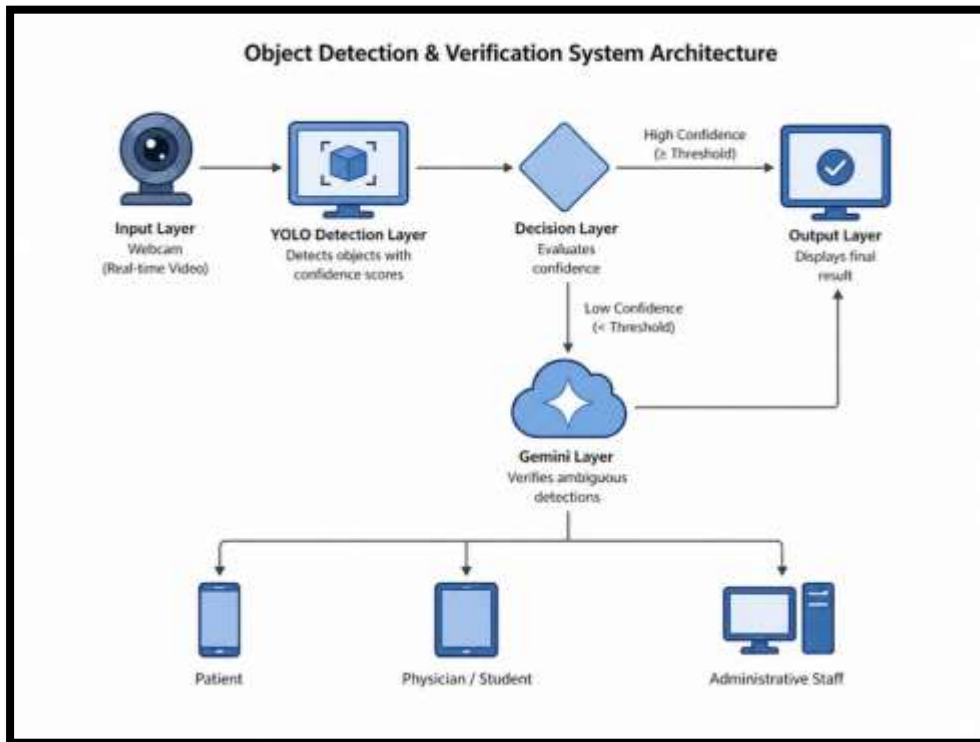
- YOLO → Fast detection
- Gemini → Intelligent reasoning

This combination enables detection of both known and unknown objects with improved accuracy.

Objectives

- Perform real-time object detection
- Improve accuracy using Generative AI
- Handle ambiguous and low-confidence detections
- Maintain high performance (FPS)
- Build a scalable AI system

3. System Architecture



Description:

The system consists of the following components:

1. Input Layer – Webcam captures real-time video
2. YOLO Detection Layer – Detects objects with confidence scores
3. Decision Layer – Evaluates confidence
4. Gemini Layer – Verifies ambiguous detections
5. Output Layer – Displays final result

4. Workflow

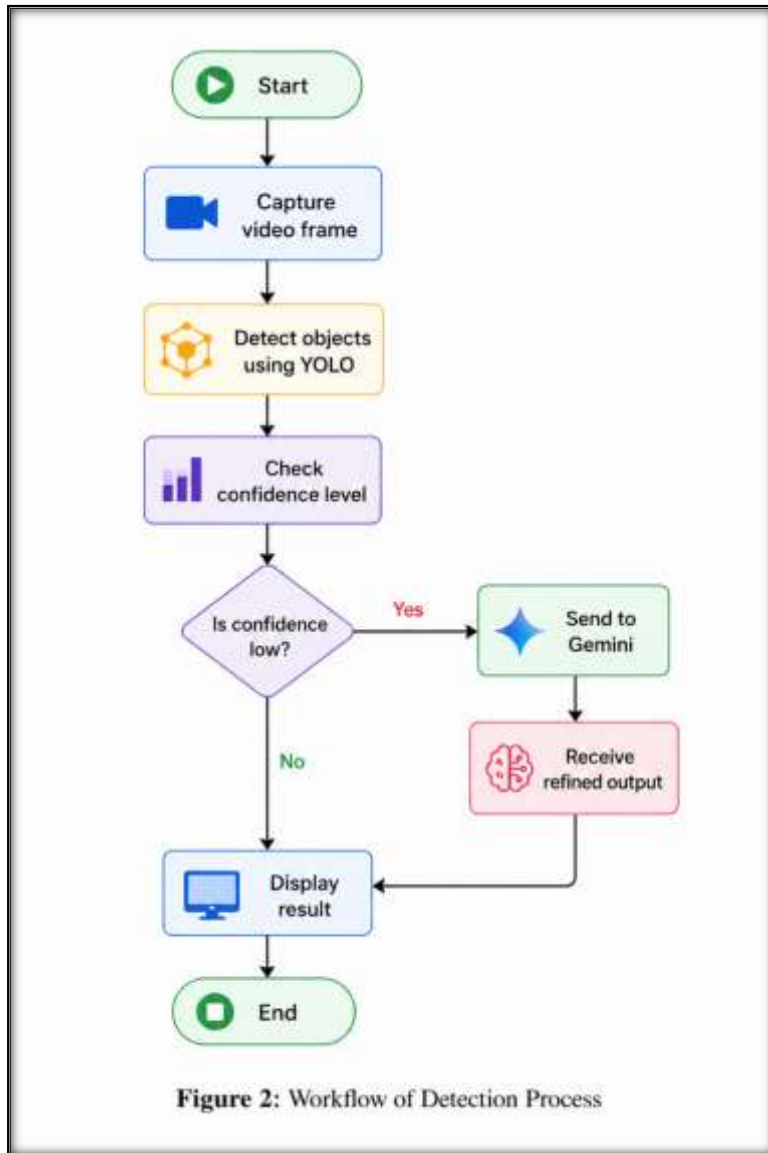


Figure 2: Workflow of Detection Process

Steps:

1. Start
2. Capture video frame
3. Detect objects using YOLO
4. Check confidence level
5. If confidence low → Send to Gemini
6. Receive refined output
7. Display result

5. Technologies Used

Category	Technology
Programming	Python
Computer Vision	OpenCV
Detection Model	YOLOv8 (YOLO-World)
Generative AI	Google Gemini 2.0 Flash
API	google-generativeai
Processing	Threading

6. Methodology (Summary)

The system uses a hybrid approach combining YOLO-based detection with Generative AI verification. The YOLO model (yolov8s-world.pt) performs real-time object detection and outputs bounding boxes, labels, and confidence scores. Based on confidence levels, the system decides whether to accept the result, ignore it, or send it for further verification.

For medium-confidence detections (15–40%), the detected object is cropped with additional padding and sent to the Gemini API, which provides a refined, context-aware description. To maintain real-time performance, Gemini API calls are executed asynchronously using a separate thread, ensuring high FPS.

7. Implementation Details / Code Snippets /Git-hub

The implementation involves three main steps:

1. Installing required dependencies such as OpenCV, Ultralytics YOLO, and Gemini API libraries
2. Configuring the Gemini API key using an environment file
3. Running the application through the main script to start real-time detection

GitHub Link - <https://github.com/MdKhan050735/Hybrid-Intelligent-Object-Detection-System-.git>

8. Key Features

- Real-time object detection using YOLO
- Open-vocabulary object recognition capability
- Intelligent verification using Generative AI
- Hybrid AI framework combining detection and reasoning
- Optimized performance using asynchronous processing

9. Future Scope

- Deployment as a mobile or web-based application
- Integration of offline AI models for reduced dependency on APIs
- Implementation of object tracking for continuous monitoring
- Integration with voice assistant systems
- Development of an advanced analytics dashboard

10. Conclusion

The Hybrid Intelligent Object Detection System successfully integrates real-time object detection with advanced Generative AI capabilities. By combining the speed of YOLO with the contextual understanding of Gemini, the system effectively addresses the limitations of traditional detection methods. This approach enhances accuracy, enables intelligent decision-making, and provides a scalable solution for real-world applications.

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